Publication repair: the original study recorded RuntimeError: Could not determine home directory.. This PDF was published afterward from retained results; no optimization was rerun.

PINN optimizer study: actual bounded derived execution

Scope and preregistration

This is a local derived study of Rathore et al., Challenges in Training PINNs: A Loss Landscape Perspective (ICML2024), not the original study or official PaperBench. Read plan.md, inputs/TASK.md and inputs/SOURCE.md for the exact raw-coordinate architecture, data, optimizer and known earlier feasibility context. All source/assets are captured before this output; the caller's actual native receipts bind their provenance. Full requested execution=True. This document reports actual completed, failed and unexecuted cells separately and does not infer scientific success from task completion.

Mathematical and numerical methods

The raw [x,t] width100 three-hidden-layer tanh/linear-output network uses float64, prescribed Xavier initialization and the exact PDE plus concatenated-condition objective. Training consumes only residual coordinates and initial/boundary conditions, never the analytic future evaluation solution. Native Adam and one-iteration strong-Wolfe L-BFGS use the public parameters; shared11k Adam prefixes are explicit. Attempt/closure/accepted-update counts differ and remain visible. A first finite zero parameter change terminates stalled, not converged; exceptions remain real incomplete/error evidence. NNCG uses the actual shifted Nyström sketch, PCG explicit residual and Armijo sufficient descent. Delta stabilizes the projected factor, not the complete Hessian.

Actual registered results

Configuration	State	Updates	Closures	HVPs	Train loss	L2RE
convection-adam-s345-lr1e-04	max_updates	41000	0	0	0.0025068	0.74342
convection-lbfgs-s345-lrnone	max_updates	41000	83632	0	0.0017897	0.61168
convection-hybrid-s345-lr1e-04	max_updates	41000	63039	0	2.7764e-05	0.051382
reaction-adam-s345-lr1e-03	max_updates	41000	0	0	5.0616e-06	0.058717
reaction-lbfgs-s345-lrnone	max_updates	41000	117998	0	1.2727e-06	0.032019
reaction-hybrid-s345-lr1e-03	max_updates	41000	84327	0	9.6766e-07	0.02593
wave-adam-s345-lr1e-03	max_updates	41000	0	0	0.027989	0.47447
wave-lbfgs-s345-lrnone	max_updates	41000	83124	0	0.019503	0.46695
wave-hybrid-s345-lr1e-03	max_updates	41000	61818	0	0.0017694	0.36804
convection-adam-s456-lr1e-04	max_updates	41000	0	0	0.0027147	0.77461
convection-lbfgs-s456-lrnone	max_updates	41000	83282	0	0.0007452	0.30021
convection-hybrid-s456-lr1e-04	max_updates	41000	61861	0	3.8088e-06	0.014923
convection-adam-s567-lr1e-04	max_updates	41000	0	0	0.0031927	0.81308
convection-lbfgs-s567-lrnone	max_updates	41000	83497	0	0.0018646	0.63254
convection-hybrid-s567-lr1e-04	max_updates	41000	65241	0	1.8691e-06	0.010019
convection-adam-s345-lr1e-03	max_updates	41000	0	0	0.00037554	0.29342
convection-hybrid-s345-lr1e-03	max_updates	41000	61987	0	1.131e-05	0.034353
convection-adam-s345-lr1e-02	max_updates	41000	0	0	0.17877	1
convection-hybrid-s345-lr1e-02	max_updates	41000	62197	0	0.0018806	0.6355
convection-gd-continuation-0.0001	max_updates	200	0	0	4.308	4.3263
convection-gd-continuation-0.001	max_updates	200	0	0	0.18334	1

Configuration	State	Updates	Closures	HVPs	Train loss	L2RE
convection-nncg-continuation-0.01	algorithm_failure	0	0	421	1.131e-05	0.034353
convection-nncg-continuation-0.1	max_updates	200	0	48276	8.7675e-06	0.030512
convection-nncg-continuation-1.0	max_updates	200	0	21702	1.049e-05	0.033746

Same-final quality

Problem / selected final	Measure	Observed	Target	Result
convection / convection-nncg-continuation-0.1	relative solution L2RE	0.03051163	<= 0.01	FAIL
convection / convection-nncg-continuation-0.1	evaluation residual RMSE	0.0006024731	<= 0.01	PASS
convection / convection-nncg-continuation-0.1	initial_value RMSE	0.000501478	<= 0.01	PASS
convection / convection-nncg-continuation-0.1	periodic_value RMSE	0.007773369	<= 0.01	PASS
reaction / reaction-hybrid-s345-lr1e-03	relative solution L2RE	0.02593015	<= 0.01	FAIL
reaction / reaction-hybrid-s345-lr1e-03	evaluation residual RMSE	0.001159852	<= 0.01	PASS
reaction / reaction-hybrid-s345-lr1e-03	initial_value RMSE	0.001135092	<= 0.01	PASS
reaction / reaction-hybrid-s345-lr1e-03	periodic_value RMSE	0.0001932285	<= 0.01	PASS
wave / wave-hybrid-s345-lr1e-03	relative solution L2RE	0.368037	<= 0.01	FAIL
wave / wave-hybrid-s345-lr1e-03	evaluation residual RMSE	0.005298477	<= 0.01	PASS
wave / wave-hybrid-s345-lr1e-03	initial_value RMSE	0.02982112	<= 0.01	FAIL
wave / wave-hybrid-s345-lr1e-03	initial_velocity RMSE	0.00177964	<= 0.01	PASS
wave / wave-hybrid-s345-lr1e-03	boundary_left RMSE	0.1023659	<= 0.01	FAIL
wave / wave-hybrid-s345-lr1e-03	boundary_right RMSE	0.1106393	<= 0.01	FAIL

Selection uses eligible terminal training loss only, exact-tie ID rule. The convection continuation-parent LR rule is separate. The pre-execution1% solution and0.01 residual/each-condition targets cannot be relaxed to fit these observations. Every component comes from one selected final, not a mixture. Small residual alone is insufficient: reaction initial error can amplify, and periodic/boundary defects are independent. No continuous-domain error certificate follows from the discrete public grid.

Diagnostics and recovery

diagnostics contains actual selected scalar/finite-difference checks, parent full/residual/constraint Ritz estimates and same-system unpreconditioned/preconditioned solves when a complete execution provides that parent. Ritz residuals mark convergence; finite Krylov results are not the complete spectrum or a certified condition number. Recovery uses new real child PIDs and the actual500->600 Adam and100->120 selected L-BFGS saved optimizer/RNG; unavailable states or child failures are reported, never replaced by replaying a prefix. Physical shared computation, logical trajectories and postprocessing costs are distinct; phase wall times overlap inherited prefixes and cannot simply be summed.

Independent representation and physical-prior controls

The same raw width100/tanh architecture can be compiled from initial/boundary facts and PDE structure using a linear fixed-feature fit. These actual network exports, fit matrices and metrics are stored under structure-controls. They are deliberately not eligible for the old randomized-optimizer selection and cannot prove that any Adam/L-BFGS/NNCG trajectory reproduced a solution. They distinguish representation capacity from the initialization/optimization restriction. Known future evaluation solutions are used only after fitting to evaluate the exported weights, never as fitting labels. No old scientific failure is relabeled as a successful optimizer result.

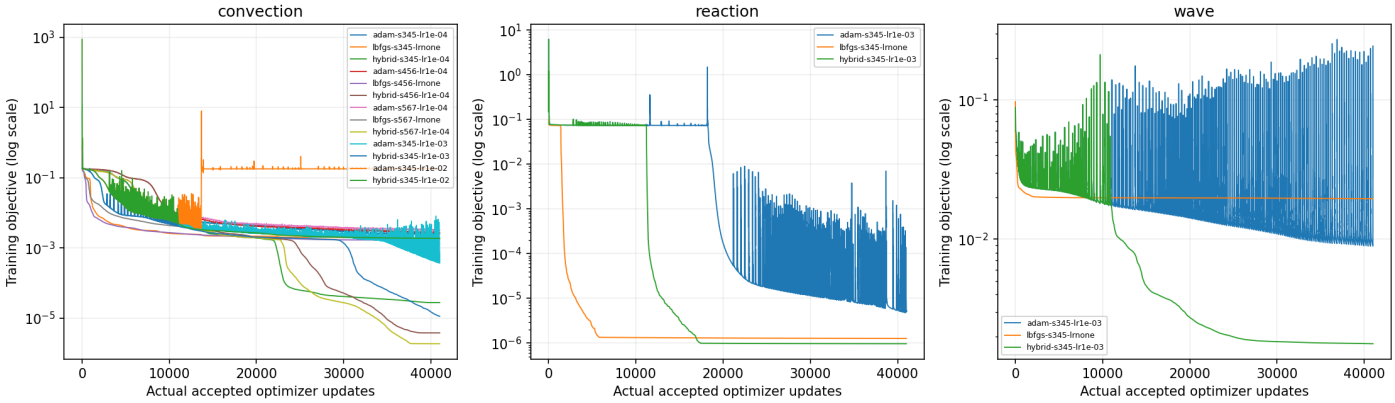
Interpretation and limits

Actual training and selected-field plots accompany this report. Cross-problem observations use seed345; only convection has paired345/456/567 and single-seed learning-rate sensitivity, so they are not19 independent statistical samples or proof of cross-problem seed robustness. All failed methods and unavailable dependencies remain in the denominator. The numerical values above, rather than an invented improvement claim, determine whether a method improved. Original Table2 convection NNCG3.63e-7 is an attributed paper value, not any result here. Current target failure is not compensated by truthful methodology. Author reference seconds and evaluator computation do not qualify effective candidate hours or statistical/professional maturity.

Reproduction and evidence

Use the captured complete source/requirements, public inputs, run_study.py command and original native execution. A fresh output retrains; reading a supplied NPZ or redrawing these figures is only re-evaluation. The actual process/argv/environment appear in study-started, phase execution, attempts, recovery and caller-owned receipts. No Provider, credentials, real user process, UI or external service is involved. The present reference is not a scored fresh OpenCorvus/Codex candidate.

Actual training trajectories



Actual selected fields and absolute errors

